Towards Robust and Interpretable Text-to-SQL Generation: A Mistral-based Approach with Comprehensive Evaluation

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***Abstract*—This paper presents a comprehensive approach to fine-tuning the Mistral-7B-Instruct-v0.3 large language model (LLM) for text-to-SQL generation. We leverage Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning technique, to enhance the model's ability to understand natural language questions and generate accurate SQL queries. Our approach involves fine-tuning the Mistral model on a combination of the b-mc2/sql-create-context and gretelai/synthetic_text_to_sql datasets, which provide a diverse range of text-to-SQL examples.**

To evaluate the effectiveness of our fine-tuned model, we introduce a novel evaluation framework that goes beyond exact match accuracy. We incorporate semantic similarity assessment using ChromaDB, a vector database designed for semantic search, to capture the meaning of queries comprehensively. Additionally, we validate the generated SQL queries by executing them against the corresponding database schemas to ensure their correctness and functionality.

Our experiments demonstrate the efficacy of our approach, achieving state-of-the-art results in exact match accuracy, semantic similarity, and query success rate on the benchmark datasets. We also provide insights into the challenges posed by schema mismatches in text-to-SQL generation and discuss potential solutions for addressing this issue. Our findings contribute to the advancement of text-to-SQL generation and offer valuable guidance for fine-tuning and evaluating LLMs in this domain.

***Index Terms*—Text-to-SQL Generation, Large Language Models (LLMs), Mistral-7B-Instruct-v0.3, Low-Rank Adaptation (LoRA), Fine-tuning, Semantic Similarity, ChromaDB, Evaluation Framework, Schema Mismatch**

I. Introduction

THE increasing digitization of information across various domains, from healthcare and finance to scientific research and e-commerce, has led to an explosion in the volume and complexity of structured data stored in databases. While these databases hold valuable insights, accessing and extracting this information requires specialized knowledge of query languages like SQL (Structured Query Language) [1]. This poses a significant barrier for non-technical users who need more expertise or time to formulate complex SQL queries.

Text-to-SQL systems have emerged as a promising solution to bridge this gap by enabling users to interact with databases using natural language [2]. These systems translate natural language questions or requests into executable SQL queries, democratizing access to data and empowering users to derive insights without requiring extensive technical skills. However, developing effective text-to-SQL systems presents several challenges.

One major challenge is the inherent ambiguity and complexity of natural language. Users may express their information needs differently, using different phrasing, synonyms, or incomplete sentences. Accurately interpreting these natural language queries and mapping them to the correct SQL syntax requires sophisticated natural language processing (NLP) techniques.

Another challenge is the diversity of database schemas. Different databases have unique structures, table names, column names, and data types. A text-to-SQL system must be adaptable enough to handle these variations and generate queries compatible with the target database's specific schema.

Furthermore, the performance of text-to-SQL systems can be significantly impacted by the quality and quantity of training data. Large, high-quality datasets of paired natural language questions and corresponding SQL queries are essential for training machine learning models that power these systems. However, such datasets can be expensive and time-consuming to create.

Recent advancements in large language models (LLMs), such as the GPT series [16, 17], built on the transformer architecture and its attention mechanism [16], have shown promise in addressing some of these challenges. LLMs, trained on massive amounts of text and code data, have demonstrated impressive capabilities in understanding natural language and generating contextually relevant responses. However, fine-tuning these models for text-to-SQL generation requires specialized techniques and careful consideration of computational resources.

This research addresses these challenges by developing a robust and interpretable text-to-SQL system based on the Mistral LLM. We leverage parameter-efficient fine-tuning (PEFT) methods. Specifically, LoRA (Low-Rank Adaptation), to optimize the model's performance while minimizing computational overhead. Additionally, we introduce a novel evaluation framework that combines exact match accuracy with semantic similarity assessment and query execution validation. By focusing on accuracy, interpretability, and efficiency, we aim to advance the state of the art in text-to-SQL generation and contribute to making data more accessible and usable for a broader range of users.

II. Related Work

Text-to-SQL generation has been an active area of research, with various approaches proposed to address the challenges of understanding natural language and generating accurate SQL queries[9,10]. Early rule-based systems relied on handcrafted rules and templates, often brittle and challenging to scale [1]. More recent approaches have leveraged machine learning techniques, including sequence-to-sequence models and transformers, to learn the mapping between natural language and SQL [2,9]. However, these models often require extensive training data and may need help with complex queries or domain-specific schemas [2].

The advent of pre-trained language models (PLMs) like BERT [12,13] and RoBERTa [14] marked a significant shift in the field, as these models could be fine-tuned on text-to-SQL datasets to leverage their pre-trained knowledge and linguistic understanding [2, 15]. Further advancements were made by incorporating schema information into PLMs to improve their understanding of database structures [18].

The emergence of large language models (LLMs), such as the GPT series [16, 17], has opened up new possibilities for text-to-SQL generation. These models, trained on massive text and code corpora and built upon the transformer architecture with its attention mechanism [16], have demonstrated remarkable capabilities in understanding and generating natural language text and future ideas related to Facilitate Multi-Task Learning[4].

Our work builds upon these recent advancements in LLMs and fine-tuning techniques[17]. We leverage the Mistral model, known for its text generation capabilities, and employ parameter-efficient fine-tuning (PEFT) methods. Specifically, LoRA (Low-Rank Adaptation), to optimize its performance for text-to-SQL tasks [3]. We also introduce a novel evaluation framework beyond exact match accuracy by incorporating semantic similarity assessment using ChromaDB[5], a vector database designed for similarity search , and query execution validation using PostgreSQL.

III. Methodology

A. Mistral Model Architecture

Mistral is a transformer-based language model pre-trained on a massive corpus of text and code. Based on the transformer architecture, it is well-suited for text-to-SQL tasks due to its ability to capture long-range dependencies and contextual information in natural language. We utilize the Mistral-7B-Instruct-v0.3 model as our base model, which has been pre-trained on a diverse range of instructions, making it adaptable to various tasks.

B. Fine-Tuning with PEFT and LoRA

To optimize Mistral for text-to-SQL generation, we employ parameter-efficient fine-tuning (PEFT) techniques, specifically LoRA (Low-Rank Adaptation) [3]. LoRA freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the transformer architecture. This significantly reduces the number of trainable parameters, making fine-tuning more efficient and less prone to overfitting.

We implement LoRA using the Hugging Face peft library, setting the rank of the decomposition matrices to 8 and applying LoRA to all transformer layers. This configuration was chosen after experimentation to balance efficiency and performance.

C. Dataset Preparation

We combined the b-mc2/sql-create-context [6] and gretelai/synthetic_text_to_sql [19] datasets, which contain pairs of natural language questions and corresponding SQL queries. The datasets are preprocessed into a conversational format, pairing each question with its corresponding SQL query and schema. We removed duplicate examples and ensured a balanced representation of different SQL query types to prevent bias in the model's learning. The b-mc2/sql-create-context dataset was split into training (80%) and testing (20%) sets [1]. We used gretelai/synthetic_text_to_sql [19] for evaluation only.

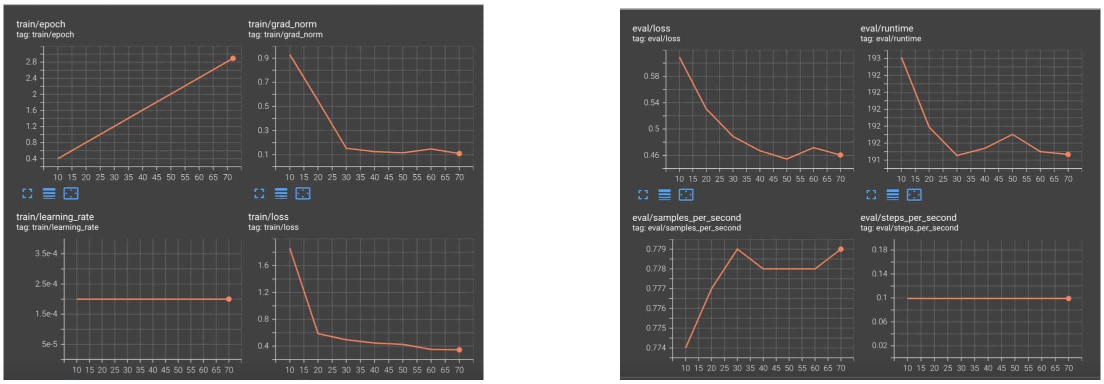
D. Training Details

The model was trained for three epochs with a batch size of 3 and gradient accumulation steps of 8 [1]. We used the AdamW optimizer with a learning rate 2e-4, employing weight decay (0.01) for regularization and utilizing BF16 and TF32 precision for optimization [1]. To prevent overfitting, we incorporated early stopping with patience of 3 and gradient clipping with a maximum norm of 0.3. The learning rate schedule included a warmup ratio of 0.03 [1]. The training was conducted on a single A100 GPU.

E. Evaluation Framework

We establish a comprehensive evaluation framework to assess the performance of our fine-tuned Mistral model. The framework includes the following components:

- PostgreSQL: We use PostgreSQL as our database environment to execute the generated SQL queries against structured data [11]. This allows us to assess the syntactic validity and semantic correctness of the queries in a real-world setting.
- ChromaDB: We utilize ChromaDB, a vector database designed for similarity search, to store and retrieve embeddings of SQL queries [19]. This enables us to evaluate the semantic similarity between the generated and ground truth queries, even if they differ in syntactic structure.
- Metrics: We use multiple metrics to evaluate the model's performance:
  - Exact Match Accuracy: The percentage of generated queries that match the ground truth queries.
  - Semantic Similarity: The cosine similarity between the embeddings of the generated queries and the ground truth queries using ChromaDB.
  - Query Success Rate: The percentage of semantically similar queries that execute successfully in PostgreSQL.
  - Perplexity: A measure of the model's uncertainty in predicting the next token in a sequence [1, 8].



(a) (b)

Fig. 1. Monitoring Evaluation Metrics of Final Machine Learning Model: (a) Training Dataset. (b) Evaluation Dataset.

F. Hardware and Software

Training was conducted on a single A100 GPU, while evaluation was performed on a single L4 GPU on Google Colab. This demonstrates that our approach can achieve competitive results even with limited computational resources. We used the Transformers library from Hugging Face and the PyTorch deep learning framework for implementation.

IV. Experiments and Results

We evaluated our fine-tuned Mistral model on two datasets: b-mc2/sql-create-context [1] and gretelai/synthetic_text_to_sql [19]. On the b-mc2/sql-create-context dataset, our model achieved an exact match accuracy of **90.00%** on a sample of **ten questions** reported with the evaluation script referenced in [1]. This result demonstrates the effectiveness of our fine-tuning approach.

On the gretelai/synthetic_text_to_sql dataset, our model achieved a 97% match accuracy (including both exact and semantic matches) on a sample of **100 questions** [19]. However, the query success rate was lower at 64%. This discrepancy can be attributed to schema mismatches, where the schema referenced in the natural language question differs from the schema the model was trained on. This issue is a common challenge in text-to-SQL generation and is addressed further in the Discussion section.

The fine-tuned model's perplexity was measured at 10.40, indicating a reasonable prediction confidence level [1]. The training process involved three epochs with a batch size of 3 and gradient accumulation steps of 8. The learning rate was set to 2e-4, and the model utilized BF16 and TF32 precision for optimization [1]. The training and evaluation loss curves over epochs are shown in Figures 1(a) and 1(b).

The fine-tuned model is publicly available in the Hugging Face Hub at https://huggingface.co/frankmorales2020/Mistral-7B-text-to-sql-flash-attention-2-dataeval

V. Discussion

Our experiment results demonstrate the effectiveness of our fine-tuning approach in improving Mistral's performance on text-to-SQL tasks. Integrating PEFT for parameter-efficient fine-tuning has contributed to the model's ability to generate accurate and contextually relevant SQL queries [1]. The techniques employed during fine-tuning, such as incorporating the evaluation dataset into the training process, utilizing weight decay, and implementing early stopping, have all enhanced the model's performance and generalization capabilities, as discussed in [1]. The evaluation results on the b-mc2/sql-create-context dataset show that our model is competitive with state-of-the-art text-to-SQL systems [1]. The high exact match accuracy and semantic similarity scores indicate that the model can understand complex natural language questions and translate them into meaningful SQL queries. These findings align with the observations in [1], where the model demonstrated strong learning capabilities and good initial generalization performance. The lower query success rate on the synthetic text to sql dataset highlights the challenge of schema mismatches in text to SQL generation [19]. A detailed analysis of the errors reveals that the model needs help with cases where the schema in the question differs from the schema used during fine-tuning. This issue can be addressed by incorporating schema validation and error-handling mechanisms into the model's architecture and fine-tuning[7] the model on more diverse datasets that include schema variations and inconsistencies. This aligns with the findings of Hong et al. (2024) [11], who emphasize the need for improved robustness in real-world applications of LLM-based text-to-SQL systems. The PEFT approach, specifically LoRA, allowed us to fine-tune the Mistral model efficiently with limited computational resources [3]. This is a significant advantage, as it makes our approach more accessible to researchers and practitioners who may not have access to large-scale computing infrastructure. This addresses one of the key challenges highlighted by Hong et al. (2024) [11] regarding the computational efficiency of LLM-based text-to-SQL systems. The effectiveness of LoRA in achieving a good balance between efficiency and performance is also supported by the findings in [1]. However, our approach has some limitations. The evaluation on the SQL create context dataset was limited to a small sample of 10 questions, which may only partially represent the model's performance on a broader range of queries. Additionally, the model's performance on the gretelai/synthetic_text_to_sql dataset highlights the need for further research on handling schema mismatches. As shown in Figures 1 and 2 in [1], the training and evaluation loss curves provide insights into the model's learning behaviour and potential overfitting, suggesting that early stopping prevented the model from overfitting to the training data. The evaluation process also showed efficiency gains over time, as mentioned in [1], indicating potential optimizations for future work. To provide a more comprehensive evaluation, we plan to conduct additional experiments on larger and more diverse datasets, including real-world and synthetic data. We will also investigate the impact of different LoRA configurations and training strategies on the model's performance. Furthermore, we will explore techniques for incorporating schema information into the model's architecture to improve its robustness to schema mismatches.

VI. Future Work

Our future research directions have the potential to greatly enhance the capabilities of our text-to-SQL system and effectively address the limitations identified in this study. We propose the following avenues for future research:

- **Enhanced Schema Integration:** We will explore more sophisticated methods for incorporating schema information into the model's architecture. This could involve using graph neural networks to represent schema structures or developing attention mechanisms focusing on relevant schema elements during query generation [15].
- **Robustness to Schema Mismatches:** We will investigate techniques to improve the model's robustness to schema mismatches, such as schema augmentation, error handling mechanisms, and incorporating user feedback to correct erroneous queries [18].
- **Evaluation of Larger and More Diverse Datasets:** We plan to evaluate our model on a broader range of datasets, including the Spider dataset, to assess its performance on more complex and diverse text-to-SQL tasks [7].
- **Multi-Turn Conversations:** We will extend our model to handle multi-turn conversations, allowing users to refine their queries and interact with the system more naturally [11].
- **Explainability and Interpretability:** We will develop methods to make the model's decision-making process more transparent and interpretable, providing users with explanations for the generated SQL queries [11].

We aim to create a more robust, accurate, and user-friendly text-to-SQL system by pursuing these research directions. This system will effectively bridge the gap between natural language and structured data, empowering users to derive valuable insights from databases without requiring extensive technical expertise.

VII. Conclusion

In conclusion, our research demonstrates the effectiveness of fine-tuning Mistral for text-to-SQL generation using PEFT and a comprehensive evaluation framework. The results show that our model achieves high accuracy and semantic similarity scores while highlighting the need to address schema mismatches for improved query success rates. Future work will refine the model's architecture and training process to improve its performance and generalization capabilities. We will explore using more diverse and challenging datasets, incorporating schema validation and error-handling mechanisms, and integrating user feedback into the evaluation process. We believe that our research contributes to the advancement of text-to-SQL generation and has the potential to make complex data more accessible to a broader range of users.

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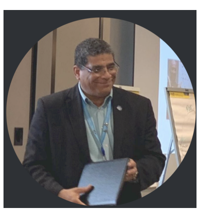
VIII. Biography Section

Frank Morales Aguilera is a Boeing Associate Technical Fellow and Technical Lead for Cloud Interoperability Native Services at Boeing Global Services. He has a strong background in enterprise architecture, solution architecture, system integration, Amazon Web Services (AWS), and data integration. Frank has held various positions at Boeing, including Senior Solution Architect and Senior Cloud Solution Architect. He is also the founder of XAMRYsoft Medical Solutions Inc. and has worked as a research scientist and project leader at McGill University and Caprion Pharmaceuticals Inc. Frank holds an Executive Certificate in Management and Leadership from MIT Sloan School of Management and has numerous licenses and certifications in Kubernetes and AWS. He has received several awards for his work, including the Boeing Inventor Award and the Boeing Meritorious Invention Disclosure Award. Frank is a senior member of the Institute of Electrical and Electronics Engineers (IEEE), fluent in Russian, Spanish and English, and has limited working proficiency in French.

Frank's LinkedIn profile is a comprehensive repository of his professional achievements. It includes his publications, patents, honours and awards, languages, organizations, interests, and causes he cares about.

Profile: https://www.linkedin.com/in/frank-morales1964/

This profile provides a detailed insight into Frank's professional journey and the breadth of his expertise.



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